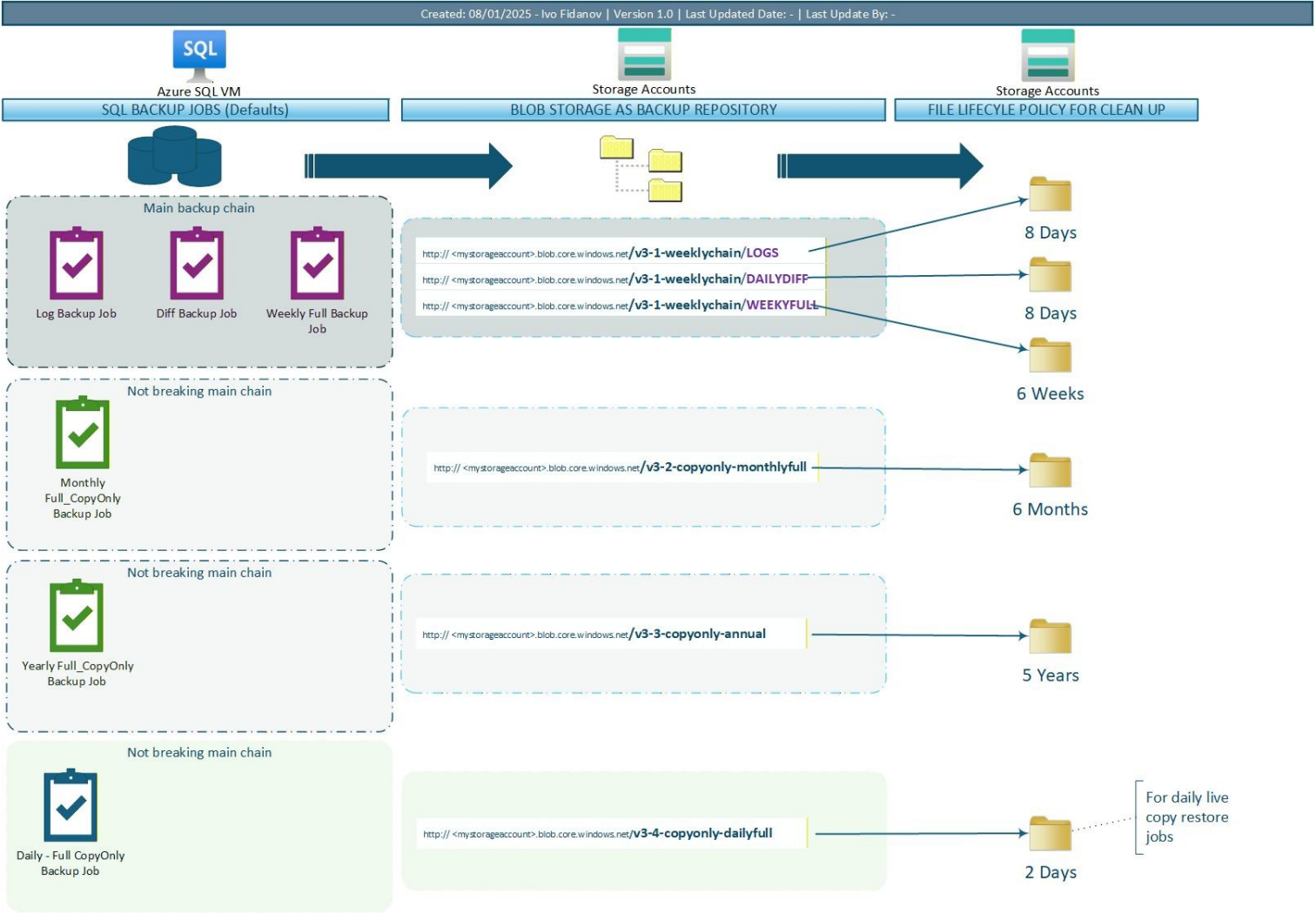


High-level Overview of PROD Microsoft SQL Server Backup Jobs

Backup Cheat Sheet

Overview

This diagram explains the overall view of the PROD Microsoft SQL Server Backups jobs and, helps anyone interested in gaining a better understanding of the data lifecycle we are maintaining for PROD backup environment.



SQL JOB STEPS

Each SQL job used for backups needs to have split steps for each database backup. Step names should be made up of database name, storage account name and container name.

This an example for step names:

- STP1: echo2-<mystorageaccount>-<container>
- STP2: geoimport-redazprodechobackups-v3-4-copyonly-dailyfull
- STP3: Echo2_DEV-redazprodechobackups-v3-4-copyonly-dailyfull
- STP4: ...

STORAGE ACCOUNTS

To meet the backup RTO and RPO targets and to support automated live restore from the latest PROD database backup to NON-PROD, backups are stored across 4 separate containers: 'v3-1-weeklychain', 'v3-2-copyonly-monthlyfull', 'copyonly-annual', and 'v3-4-copyonly-dailyfull'.

Following tables helps you to adjust repositories(containers) in storage accounts and to set lifecycle target for the repositories:

Storage Account	To Set Lifecycle Policy	Notes	Sql1	Sql2
http://<mystorageaccount>.blob.core.windows.net/v3-1-weeklychain/logs	8 days		x	
http://<mystorageaccount>.blob.core.windows.net/v3-1-weeklychain/dailydiff	8 days		x	
http://<mystorageaccount>.blob.core.windows.net/v3-1-weeklychain/weeklyfull	6 weeks		x	
http://<mystorageaccount>.blob.core.windows.net/v3-2-copyonly-monthlyfull	12 months			x
http://<mystorageaccount>.blob.core.windows.net/v3-3-copyonly-annual	3 years			x
http://<mystorageaccount>.blob.core.windows.net/v3-4-copyonly-dailyfull	2 days	used for daily restore automations		x

SQL JOB NAMES

To fill with, the scheduled day `@EveryFirstFriday` and the scheduled time `11:PM` please look the (will create in sql maintenance windows documentation @Leslie) document. The '**dbname**' field is optional and should only be populated if you want to indicate that a specific database is being backed up within that job."

- DBOps - BackupV3-1 - Log - @Every15mins - Dur:--mins~
- DBOps - BackupV3-1 - Dailydiff - @Everyday 22:00 PM - Dur:--mins~
- DBOps - BackupV3-1 - Weeklyfull - @Saturday 19:00 PM - Dur:--mins~
- DBOps - BackupV3-2 - CopyOnly monthlyfull- @Sunday 19:00PM - Dur:--mins~
- DBOps - BackupV3-3 - CopyOnly annual- @Saturday 19:00PM - Dur:--mins~
- DBOps - BackupV3-4 - CopyOnly dailyfull- @Sunday 23:00 PM - Dur:--mins~

SQL JOBS

1-LOG BACKUP

```
EXECUTE dpAudit.dbo.DatabaseBackup
@Databases = '<DBNAME>',
@URL = 'http:// <mystorageaccount>.blob.core.windows.net/v3-1-weeklychain/logs',
@BackupType = 'LOG',
@Compress = 'Y',
@Verify = 'Y',
@Checksum = 'N',
@OverrideBackupPreference = 'Y',
@LogToTable = 'Y',
@DirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}',
@AvailabilityGroupDirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}'
```

2-DAILY DIFF BACKUP

```
EXECUTE dpAudit.dbo.DatabaseBackup
@Databases = '<DBNAME>',
@URL = 'http:// <mystorageaccount>.blob.core.windows.net/v3-1-weeklychain/dailydiff',
@BackupType = 'DIFF',
@Compress = 'Y',
@Checksum = 'N',
@Verify = 'Y',
@BufferCount = 50,
@MaxTransferSize = 4194304,
@NumberOfFiles = 1,
@LogToTable = 'Y',
@DirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}',
@AvailabilityGroupDirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}'
```

3-WEEKLY FULL BACKUP

```
EXECUTE dpAudit.dbo.DatabaseBackup
@Databases = '<DBNAME>',
@URL = 'http:// <mystorageaccount>.blob.core.windows.net/v3-1-weeklychain/weeklyfull',
```

```
@BackupType = 'FULL',
@Compress = 'Y',
@Checksum = 'N',
@BufferCount = 50,
@MaxTransferSize = 4194304,
@NumberOfFiles = 1,
@LogToTable = 'Y',
@DirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}',
@AvailabilityGroupDirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}'
```

4-COPY ONLY MONTHLY FULL

```
EXECUTE dpAudit.dbo.DatabaseBackup
@Databases = '<DBNAME>',
@URL = 'http:// <mystorageaccount>.blob.core.windows.net/v3-2-copyonly-monthlyfull',
@BackupType = 'FULL',
@Compress = 'Y',
@Checksum = 'N',
@Verify = 'Y',
@BufferCount = 50,
@MaxTransferSize = 4194304,
@NumberOfFiles = 1,
@CopyOnly = 'Y',
@LogToTable = 'Y',
@DirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}',
@AvailabilityGroupDirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}'
```

5-COPY ONLY ANNUAL FULL

```
EXECUTE dpAudit.dbo.DatabaseBackup
@Databases = '<DBNAME>',
@URL = 'http:// <mystorageaccount>.blob.core.windows.net/v3-3-copyonly-annual',
@BackupType = 'FULL',
@Compress = 'Y',
@Checksum = 'N',
@Verify = 'Y',
@BufferCount = 50,
@MaxTransferSize = 4194304,
@NumberOfFiles = 1,
@CopyOnly = 'Y',
@LogToTable = 'Y',
@DirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}',
@AvailabilityGroupDirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}'
```

6-COPY ONLY DAILY FULL

```
EXECUTE dpAudit.dbo.DatabaseBackup
@Databases = '<DBNAME>',
@URL = 'http:// <mystorageaccount>.blob.core.windows.net/v3-4-copyonly-dailyfull',
@BackupType = 'FULL',
@Compress = 'Y',
@Checksum = 'N',
@Verify = 'Y',
@BufferCount = 50,
@MaxTransferSize = 4194304,
@NumberOfFiles = 1,
@CopyOnly = 'Y',
@LogToTable = 'Y',
@DirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}',
@AvailabilityGroupDirectoryStructure = '{DatabaseName}{DirectorySeparator}{BackupType}_{Partial}_{CopyOnly}'
```

LIFECYLE POLICY

```
{
  "rules": [
    {
      "name": "delete-weeklychain-logs",
      "enabled": true,
      "type": "Lifecycle",
      "definition": {
        "filters": {
          "blobTypes": ["blockBlob"],
          "prefixMatch": ["v3-1-weeklychain/logs"]
        },
        "actions": {
          "baseBlob": {
            "delete": {
              "daysAfterCreationGreaterThan": 8
            }
          }
        }
      }
    }
  ]
}
```

```

    }
  },
  {
    "name": "delete-weeklychain-dailydiff",
    "enabled": true,
    "type": "Lifecycle",
    "definition": {
      "filters": {
        "blobTypes": ["blockBlob"],
        "prefixMatch": ["v3-1-weeklychain/dailydiff"]
      },
      "actions": {
        "baseBlob": {
          "delete": {
            "daysAfterCreationGreaterThan": 8
          }
        }
      }
    }
  },
  {
    "name": "delete-weeklychain-weeklyfull",
    "enabled": true,
    "type": "Lifecycle",
    "definition": {
      "filters": {
        "blobTypes": ["blockBlob"],
        "prefixMatch": ["v3-1-weeklychain/weeklyfull"]
      },
      "actions": {
        "baseBlob": {
          "delete": {
            "daysAfterCreationGreaterThan": 42
          }
        }
      }
    }
  },
  {
    "name": "delete-copyonly-monthlyfull",
    "enabled": true,
    "type": "Lifecycle",
    "definition": {
      "filters": {
        "blobTypes": ["blockBlob"],
        "prefixMatch": ["v3-2-copyonly-monthlyfull"]
      },
      "actions": {
        "baseBlob": {
          "tierToCool": {
            "daysAfterModificationGreaterThan": 30
          },
          "tierToCold": {
            "daysAfterModificationGreaterThan": 90
          },
          "tierToArchive": {
            "daysAfterModificationGreaterThan": 180
          },
          "delete": {
            "daysAfterCreationGreaterThan": 180
          }
        }
      }
    }
  },
  {
    "name": "delete-copyonly-annual",
    "enabled": true,
    "type": "Lifecycle",
    "definition": {
      "filters": {
        "blobTypes": ["blockBlob"],
        "prefixMatch": ["v3-3-copyonly-annual"]
      },
      "actions": {
        "baseBlob": {
          "tierToCool": {
            "daysAfterModificationGreaterThan": 30
          },
          "tierToCold": {
            "daysAfterModificationGreaterThan": 90
          },
          "tierToArchive": {

```

```
        "daysAfterModificationGreaterThan": 365
      },
      "delete": {
        "daysAfterCreationGreaterThan": 1825
      }
    }
  },
  {
    "name": "delete-copyonly-dailyfull",
    "enabled": true,
    "type": "Lifecycle",
    "definition": {
      "filters": {
        "blobTypes": ["blockBlob"],
        "prefixMatch": ["v3-4-copyonly-dailyfull"]
      },
      "actions": {
        "baseBlob": {
          "delete": {
            "daysAfterCreationGreaterThan": 2
          }
        }
      }
    }
  }
]
}
```